Chase Oswalt

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Website: http://chaseogit.github.io/

EDUCATION

University of Central Florida | Orlando, FL | 2022-2026
Bachelor of Science in Information Technology | Conferred: August 2026

- GPA: 3.53 / 4.00 | Honors: Dean's List
- Relevant Coursework: Enterprise Computing, Database Systems, Object-Oriented Programming, Systems Software, System Administration & Maintenance, Human-Computer Interaction (HCI)

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript (ES6+), C, C++, C#, SQL, PowerShell, HTML5, CSS3, GLSL
- **Frameworks & Web:** React.js, Node.js, Express.js, Java Servlets, JSP, JDBC, REST APIs, Vite, Nginx
- **Cloud & Systems:** AWS (Cloud Architecture), DigitalOcean, Linux, Windows Server 2022, Active Directory, Docker (Basics), Oracle VirtualBox, Git/GitHub, GitHub Actions CI/CD
- **Databases & Tools:** MongoDB, MySQL, Mongoose, Jest, Supertest, Postman, Godot Engine
- **Certifications:** AWS Academy Graduate - AWS Academy Cloud Architecting

TECHNICAL PROJECTS

UCF Technology Lending Web Platform | React, Node.js, Express, MongoDB, DigitalOcean, GitHub Actions

- Led a 4-person agile team to engineer a responsive MERN web application replacing a legacy university hardware lending system with a modern, role-aware (Student/Faculty/Admin) interface.
- Architected end-to-end authentication utilizing Google OAuth 2.0, JWT middleware, and bcrypt encryption, coupled with the Resend HTTP API for automated verification emails.
- Integrated ZXing webcam barcode scanning and external data APIs (DummyJSON, Wikipedia OpenSearch) to automate cataloging and accelerate equipment check-in/out workflows.
- Designed a MongoDB/Mongoose schema supporting Role-Based Access Control (RBAC), real-time inventory state management, and automated late-fine calculation algorithms.

- Built a CI/CD pipeline via GitHub Actions deploying to a DigitalOcean Droplet, configuring Nginx (Gzip, HTTP/2, static caching) and Jest unit tests to achieve a >95% Google Lighthouse score.
  Code: https://github.com/ChaseOGit/ucf-tech-lending-platform

Enterprise Distributed Three-Tier Web Application | Java, JSP, Servlets, Apache Tomcat, MySQL, JDBC

- Developed a scalable three-tier client-server web application deployed on an Apache Tomcat container interacting with a backend MySQL relational database.
- Implemented strict Role-Based Access Control (RBAC) to enforce granular read/write/audit permissions across four distinct user tiers (Root, Client, Data Entry, Accountant).
- Engineered server-side business logic using JDBC PreparedStatement and CallableStatement to trigger remote stored procedures, ensuring data integrity and preventing SQL injection vulnerabilities.
- Designed dynamic JSP frontends interfacing with transactional database operations enforcing foreign key referential integrity.
  Code: https://github.com/ChaseOGit/enterprise-three-tier-app

Enterprise Virtual Infrastructure & Automation Pipeline | Windows Server 2022, Active Directory, PowerShell, Python

- Architected an isolated enterprise virtual network supporting Windows Server 2022 domain controllers and multi-version Windows clients via Oracle VirtualBox.
- Configured Active Directory Domain Services (AD DS), implementing Group Policy Objects (GPOs), Organizational Units (OUs), DNS/DHCP server roles, and RBAC user provisioning.
- Authored automation scripts in PowerShell and Python to streamline routine server maintenance, health monitoring, and bulk user onboarding operations.

2D Physics Arena Combat Game | Godot Engine, C#, GLSL, Javelin Physics

- Engineered an extensible modular game architecture in Godot utilizing C# for core state-machine logic, collision responses, and local multi-gamepad input handling.
- Programmed custom GLSL shaders to render real-time post-processing visual effects and dynamic lighting passes.
- Built an internal CSV-driven data pipeline to dynamically load and balance character statistics, item attributes, and asset configurations at runtime.
  Gameplay Video Demo: https://youtu.be/4_mdrE4RQnI

EXTRACURRICULAR & LEADERSHIP

- **AI@UCF (Artificial Intelligence Club)** | Active Member: Analyzed deep learning architectures, transformer models, and neural network fine-tuning methodologies for LLMs.
- **GameDev Knights** | Active Member: Collaborated in game jams and attended workshops focused on game design patterns, memory optimization, and physics engines.

AWARDS & SCHOLARSHIPS

- **Elbit America STEM Scholarship** - Merit-based award recognized for academic excellence in computing and technology.
- **Florida Bright Futures Medallion Scholarship** - State academic merit scholarship covering undergraduate tuition.